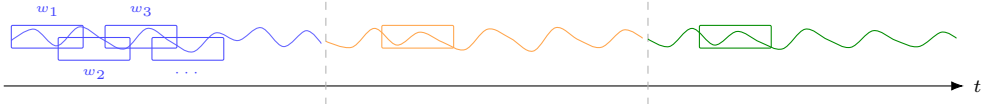


Subject $s \in \mathcal{S}$: vehicle signals $\mathbf{x}(t)$



Feature Extraction & Window Labeling: $(\mathbf{x}_w, y_w), \mathbf{x}_w \in \mathbb{R}^p, y_w \in \{0, 1\} \Rightarrow \mathcal{D}_s = \{(\mathbf{x}_w, y_w)\}_{w=1}^{n_s}$

Train pool: $\mathcal{D}_{\text{pool}} = \bigcup_{s \in \mathcal{S}_{\text{tr}}} \mathcal{D}_s^{\text{tr}}$

Val pool: $\mathcal{D}_{\text{val}} = \bigcup_{s \in \mathcal{S}_{\text{val}}} \mathcal{D}_s^{\text{val}}$

Test pool: $\mathcal{D}_{\text{test}} = \bigcup_{s \in \mathcal{S}_{\text{te}}} \mathcal{D}_s^{\text{te}}$

Classifier Training $f: \mathbb{R}^p \rightarrow [0, 1]$

Threshold Optimization: $\hat{\tau}$

Evaluation